Application Program Development

Ch 01: Introduction to Computers and Programming

Derek Harter

Department of Computer Science
East Texas A&M University

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Introduction to Computers and Programming

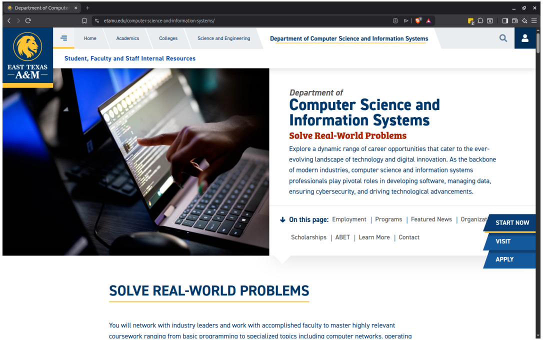

Figure 1: A web browser application program

- Computers can be programmed
  - Designed to do any job that a program tells them to
- **Program**: set of instructions that a computer follows to perform a task.
  - Commonly referred to as **Software**
- **Programmer**: person who can design, create and test computer programs
  - Also known as a **software developer**

Computer Hardware Components

- **Hardware**: The physical devices that make up a computer
  - Computer is a system composed of several components that all work together

Typical major components:

- Central processing unit (CPU)
- Main memory (RAM)
- Secondary storage devices (hard drives, SSD)
- Input and output devices

Typical Components of a Computer System

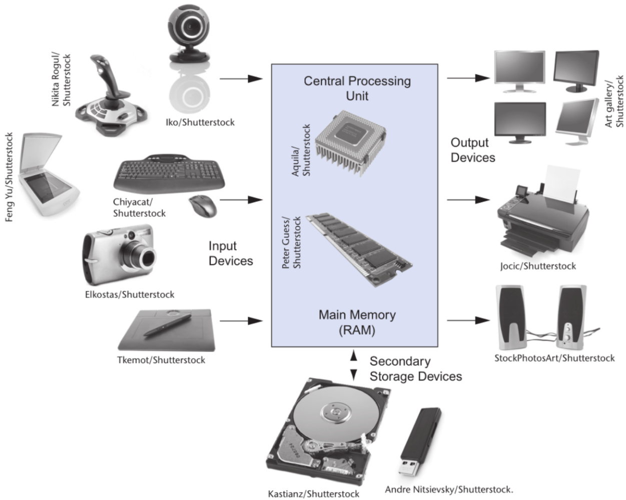

Figure 2: Typical components of a computer system (Gaddis, 2021, pg.25)

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Computer Software

- Everything a computer does is controlled by software
- **Application software**: programs that make computer useful for every day tasks
  - Word processing, email, web browsers, programming editors, photo editing
- **System software**: programs that control and manage basic operations and resources of a computer
  - **Operating system**: controls operations of hardware components
  - **Utility programs**: perform specific tasks to enhance computer operation
  - **Software development tools**: used to create, modify and test software

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How Computers Store Data

- All data in a computer is stored in sequences of 0s and 1s
- **Byte**: just enough memory to store letter or small number
  - Divided into eight bits
  - **Bit**: electrical component that can hold positive or negative charge, on/off switch
  - The on/off pattern of bits in a byte represents data stored in the byte

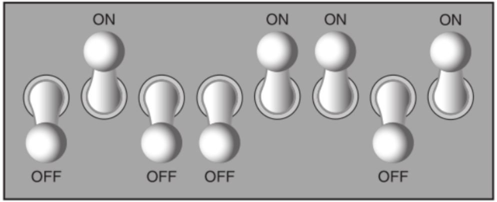

Figure 3: Think of a byte as eight switches (Gaddis, 2021, pg.30)

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Storing Numbers

- Computers use binary numbering system
- $0100 \;\; 1101 = 2^6 + 2^3 + 2^2 + 2^0 = 64 + 8 + 4 + 1 = 77$
- Byte size limits are 0 and 255
  - $0 = 0000 \;\; 0000$ all bits off; $255 = 1111 \;\; 1111 =$ all bits on
  - To store larger number, use several bytes

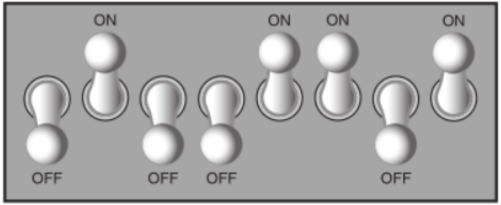

Figure 4: The number 77 stored in a byte (Gaddis, 2021, pg.30)

Storing Characters

- Data stored in computer must be stored as binary number
- Characters are converted to numeric code, interpreted as characters
  - Basic character encoding scheme is ASCII Appendix C (7 bits or 128 characters)
  - Unicode is current standard
    - Compatible with but extends ASCII
    - Can represent many character sets, for many other languages than English

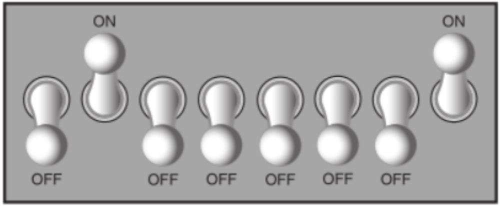

Figure 5: The letter A ASCII encoding in a byte (Gaddis, 2021, pg.30)

Advanced Number Storage

- To store negative numbers and real (float) numbers, computers use binary numbering encoding schemes
  - Negative signed numbers encoded using two's complement
  - Real numbers encoded using floating-point notation IEEE 754

Other Types of Data

- **Digital** describes any device that stores data as binary numbers
- Digital images are composed of pixels
  - Each pixel converted to a binary number representing pixel color
  - Video is a sequence of digital images
- Digital music and audio is composed of digitized sound samples
  - Each sound level sample is converted to a binary number

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References

Gaddis, T. (2021). *Starting out with python* (5th). Pearson.

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